

DES

Steps to Solve DES.

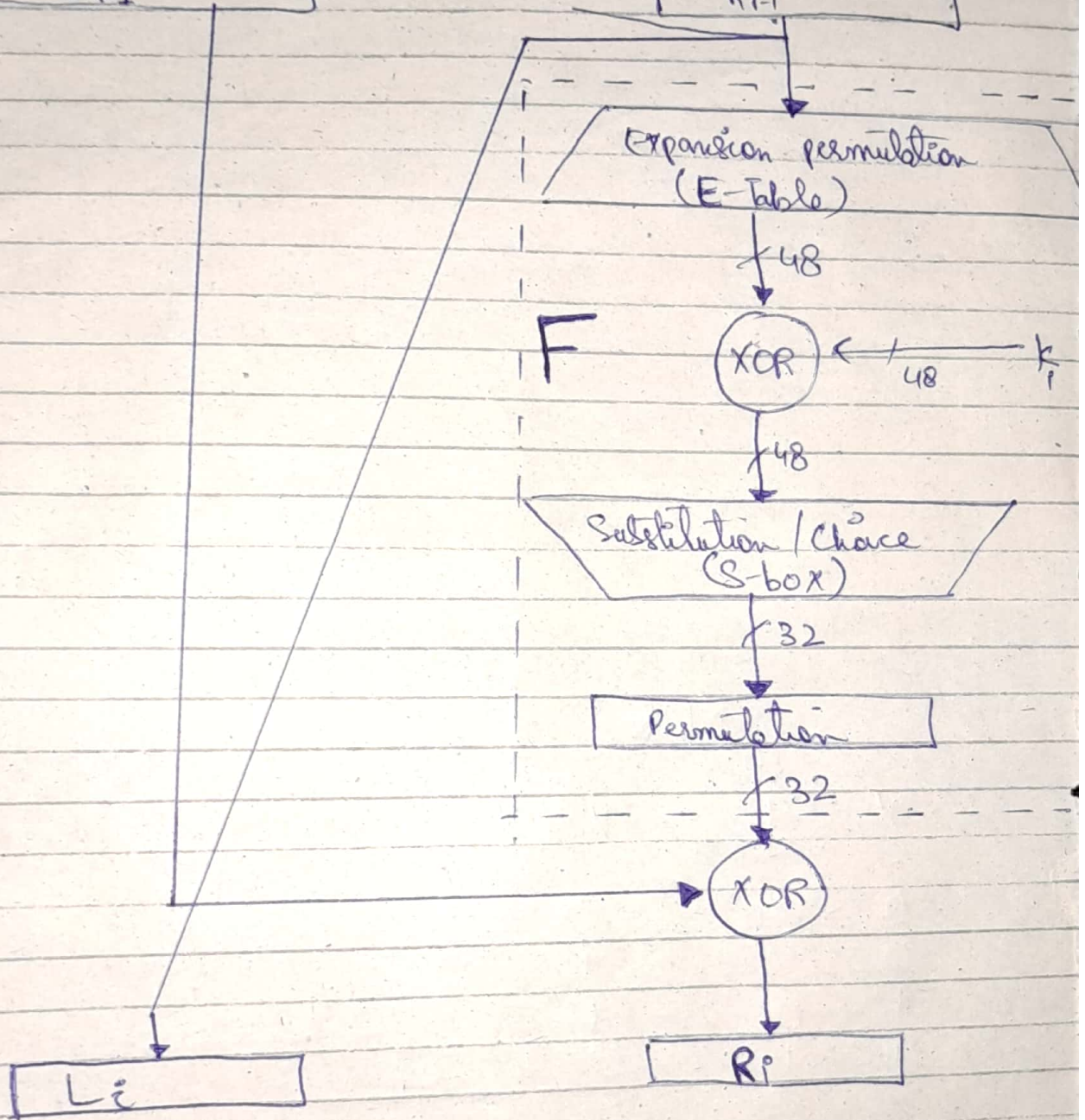
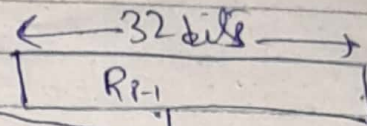
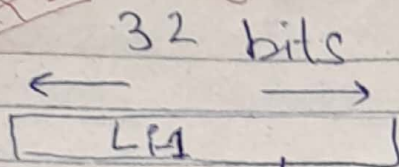
40

- 1) Calculate character of PT.
- 2) Convert it into ASCII
- 3) Convert into Binary
- 4) If exceeds 64 bit make another block
apply padding (zero padding, space padding, null padding etc)
if not exceeds skip this step.
- 5) Input (Block) $\rightarrow$ IP (Initial permutation)
and produce final permutation.
- 6) Key $\rightarrow$ PC-1 = C and D
- 7) Apply Circular Shift and PC-2

XOR

X	Y	$X \oplus Y$
0	0	0
0	1	1
1	0	1
1	1	0

Q1 diagram



1

Q3

DES

P.T: John Wick!

ASCII value of P.T = J → 74, O → 111, h → 104
 n → 110, space → 32, W → 87
 i → 105, c → 99, k → 107,
 ! = 33

P.T	ASCII	Binary	As we have 10 characters in P.T, first we convert it into ASCII value, then convert it into binary, because DES uses bits to encrypt, that makes it a bit wise encrypting algorithm. When we try to make it into block of 64, it exceeds the
1 J	74	01001010	
2 O	111	01101111	
3 h	104	01101000	
4 n	110	01101110	
5 Space	32	00100000	
6 W	87	01010111	
7 i	105	01101001	
8 c	99	01100011	
9 k	107	01101011	
10 !	33	00100001	

block, and to perform complete encryption we will make another block of 64 that consist of $80 - 64 = 16$ bit and we apply padding (zero padding, ISO/IEC 7816-4 padding, ISO/IEC 10126 padding etc)

~~we~~ ~~used~~ to make a block of 64 bit and apply DES to both blocks 1-by-1.

Now lets Start:-

J 01001010
 O 01101111
 h 01101000
 n 01101110

Space 00100000
 W 01010111
 i 01101001
 c 01100011

k 01101011
 ! 00100001

②

Block 1

01001010
01101111
01101000
01101110
00100000
01010111
01101001
01100011

Block 2

01101011
00100001
00000000
00000000
00000000
00000000
00000000
00000000
00000000

Now we have two blocks of 64-bits
we will apply DES one-by-one on both
blocks.

Step 5

01001010
01101111
01101000
01101110
00100000
01010111
01101001
01100011

58 50 42 34 26 18 10 02
60 52 44 36 28 20 12 04
62 54 46 38 30 22 14 06
64 56 48 40 32 24 16 08
57 49 41 33 25 17 09 01
59 51 43 35 27 19 11 03
61 53 45 37 29 21 13 05
63 55 47 39 31 23 15 07

Block 1

IP table.

By using the combinations given in helping
Material named Final Permutation, we get,

(3)

we got the table: $PT \rightarrow IP$

1	1	1	0	1	1	1	1
0	0	1	0	0	0	0	0
0	0	1	0	1	0	1	0
1	1	1	0	0	0	1	0
0	0	0	0	0	0	0	0
1	1	0	1	1	1	1	0
0	1	0	0	1	1	1	1
1	0	1	0	1	0	1	1

Initial-Permutation

Step 6 This process will convert key into 56 bit from 64 bit.

0	0	1	1	0	1	0	0	57	49	41	33	25	17	9
0	0	1	0	1	1	0	1	01	58	50	42	34	26	18
1	0	1	1	0	1	0	1	10	2	59	51	43	35	27
1	0	1	0	1	0	0	0	19	11	3	60	52	44	36
0	0	0	1	1	1	0	1	63	55	47	39	31	23	15
1	1	0	1	1	0	1	1	7	62	54	46	38	30	22
1	0	0	1	0	0	0	0	14	6	61	53	45	37	29
0	0	0	0	0	1	0	0	21	13	5	28	20	12	4

Block 1

PC-1

0 1 1 0 1 1 0
0 0 0 1 0 0 0
0 0 0 0 0 0 1
1 1 1 0 1 1 1

0 0 1 0 0 0 0
0 1 0 0 1 0 1
1 1 0 0 1 1 1
0 1 0 0 1 0 1

C

D

(4)

Now our key is converted into 56 bits.

Step #7

Now we will apply ~~left~~ circular shift according to our round level.

1. Left circular Shift

RA#	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Shift	1	1	2	2	2	2	2	2	1	2	2	2	2	2	2	1

C

0	1	1	0	1	1	0										
0	0	0	1	0	0	0					1	1	0	1	1	0
0	0	0	0	0	0	0	1				0	0	1	0	0	0
1	1	1	0	1	1	1					0	0	0	0	0	1

D

0	0	1	0	0	0	0										
0	1	0	0	1	0	1					0	1	0	0	0	0
1	1	0	0	1	1	1					1	0	0	1	0	1
0	1	0	0	1	0	1					1	0	0	1	0	1

C_s

0	1	1	0	1	1	0										
0	0	0	1	0	0	0					0	0	1	0	0	0
0	0	0	0	0	0	0	1				0	0	0	0	0	1
1	1	0	1	1	1	0					1	1	0	1	1	0

2) PC-2

1	1	0	1	1	0	0				14	17	11	24	1	5
0	0	1	0	0	0	0				3	28	15	6	21	10
0	0	0	0	0	1	1				23	19	12	4	26	8
1	1	0	1	1	1	0				16	7	27	20	13	2
0	1	0	0	0	0	0				41	52	31	37	47	55
1	0	0	1	0	1	1				30	40	51	45	33	48
1	0	0	1	1	1	0				44	49	39	56	34	53
1	0	0	1	0	1	0				46	42	50	36	29	32

C+D

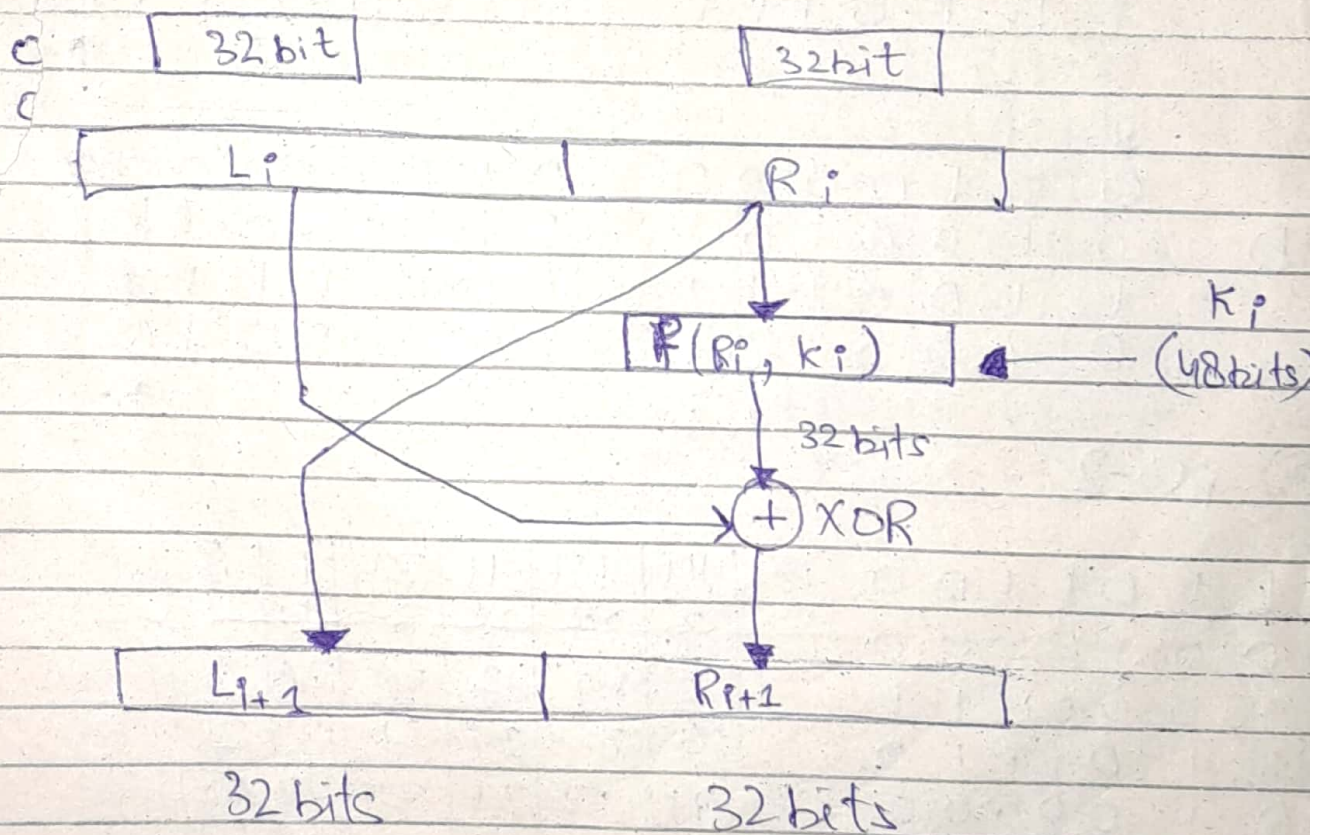
8x6
8

Key $\rightarrow$ PC-2 = below Table.

0	0	0	0	1	1
0	0	0	0	1	1
1	0	0	1	1	0
0	0	1	1	0	1
1	0	0	0	1	1
1	0	0	0	0	1
0	0	1	0	0	1
1	1	1	1	0	0

Key

Now our key is converted into 48 bits we will pass these key through Substitution function F - (8 S-boxes) producing 32-bit output.



$\therefore i = \text{round numbers}$



Step-8 Applying E-box on the right 32 bits or (second half) $\rightarrow$ of IP

1	2	3	4	5	6	7	8
0	0	0	0	0	0	0	0
9	10	11	12	13	14	15	16
0	1	0	1	1	1	1	0
17	18	19	20	21	22	23	24
1	0	1	0	1	0	1	1
25	26	27	28	29	30	31	32

32	1	2	3	4	5
4	5	6	7	8	9
8	9	10	11	12	13
12	13	14	15	16	17
16	17	18	19	20	21
20	21	22	23	24	25
24	25	26	27	28	29
28	29	30	31	32	1

1	0	0	0	0	0	0	0
0	0	0	0	0	1	0	0
0	1	1	0	1	1	0	0
1	1	1	1	0	0	0	0
0	0	1	0	0	1	0	0
0	1	1	1	1	1	1	1
1	1	0	1	0	1	0	1
0	1	0	1	1	0	1	0

$\rightarrow$ 48 bit

By repeating the last two bit in previous row we can make 48 bits from 32 bits input.

Step-9 Input $\oplus$ Key.

1	0	0	0	0	0	0	0
0	0	0	0	0	0	1	1
0	1	1	0	1	1	1	0
1	1	1	1	0	0	0	0
0	0	1	0	0	1	0	0
0	1	1	1	1	1	1	1
1	1	0	1	0	1	0	1
0	1	0	1	1	0	1	0

$\oplus$

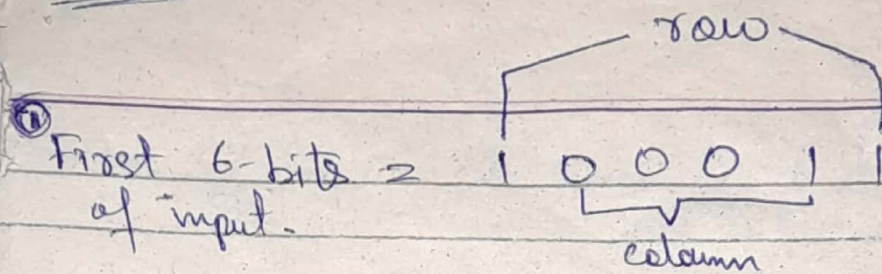
0	0	0	0	1	1	1	0
0	0	0	0	1	1	0	0
1	0	0	1	1	0	1	0
0	0	1	1	0	1	0	1
1	0	0	0	1	1	0	1
0	0	1	0	0	1	0	0
1	1	1	1	0	0	1	0
1	0	1	0	1	0	1	0

=

1	1	0	0	0	1	1	0
1	0	1	0	1	0	1	0
1	1	1	1	1	0	1	0
1	1	1	1	0	0	1	0
1	0	1	0	1	0	1	0

S-BOX (7)

Step 10 S box



row = 11 = 3 output = 12
column = 0001 = 1 binary = 1100

② 000010

row = 00 = 0 output = 01
col = 0001 = 1 binary = 0001

③ 111101

row = 11 = 3 output = 02
col = 1110 = 14 binary = 10010

④ 110001

row = 11 = 3 output = 09
col = 1000 = 8 binary = 1001

⑤ 101010

row = 10 = 2 output = 13
col = 0101 = 5 binary = 1101

⑥ 111110

row = 10 = 2 output = 06
col = 1111 = 15 binary = 0110

⑦ 1 1 1 1 0 0

row = 10 = 2 output = 09
col = 1110 = 4 binary = 1001

⑧ 1 0 1 0 1 0

row = 10 = 2 output = 12
col = 0101 = 5 binary = 1100

After applying S-Box our output becomes
32-bit (4x8)

1 0 0 0 1 1	1 1 0 0
0 0 0 0 1 0	0 0 0 1
1 1 1 1 0 1	0 0 1 0
1 1 0 0 0 1	1 0 0 1
1 0 1 0 1 0	1 1 0 1
1 1 1 1 1 0	0 1 1 0
1 1 1 1 0 0	1 0 0 1
1 0 1 0 1 0	1 1 0 0

48 bits → 32 bits

⑨

Step 11 Permutation (Straight Permutation Box)

4	1	1	0	0	16	0	7	20	21	29	12	28	17
8	0	0	0	1	01	15	23	26	05	18	31	10	
12	0	0	1	0	02	08	24	14	32	27	03	09	
16	1	0	0	1	19	13	30	06	22	11	04	25	
20	1	1	0	1									
24	0	1	1	0									
28	1	0	0	1									
32	1	1	0	0									

New 32-bit (8x4)

1	0	1	0	1	0	1	1
1	0	1	0	0	1	0	0
1	1	0	0	0	0	0	0
0	1	1	0	1	1	0	1

Step 12 XOR with first half of the bits we get from Step 5 (IP)

10101011	11101111
10100100	00100000
11000000	00101010
01101101	11100010

Output.

0	1	0	0	0	1	0	0
1	0	0	0	0	1	0	0
1	1	1	0	1	0	1	0
1	0	0	0	1	1	1	1

New Second half

New R_i

(10)

New block-1 after complete Round 1:

00000000

11011110

01001111

10101011

01000100

10000100

11101010

10001111

ASCII Symbol

222

Nul

79

p

171

O

68

<<

132

D

234

"

143

^

Blank ()

11

(Zero Padding is used)

Block 2

0 1 1 0 1 0 1 1	58	50	42	34	26	18	10	02
0 0 1 0 0 0 0 1	60	52	44	36	28	20	12	04
0 0 0 0 0 0 0 0	62	54	46	38	30	22	14	06
0 0 0 0 0 0 0 0	64	56	48	40	32	24	16	08
0 0 0 0 0 0 0 0	57	49	41	33	25	17	09	01
0 0 0 0 0 0 0 0	59	51	43	35	27	19	11	03
0 0 0 0 0 0 0 0	61	53	45	37	29	21	13	05
0 0 0 0 0 0 0 0	63	55	47	39	31	23	15	07

First half	0	0	0	0	0	0	0	1
	0	0	0	0	0	0	0	0
	0	0	0	0	0	0	0	0
	0	0	0	0	0	0	1	1
Second half	0	0	0	0	0	0	0	0
	0	0	0	0	0	0	1	1
	0	0	0	0	0	0	0	1
	0	0	0	0	0	0	0	1

Initial Permutation

ep8 E-Box

0 0 0 0 0 0 0 0
 0 0 0 0 0 0 1 1
 0 0 0 0 0 0 0 1
 0 0 0 0 0 0 0 1

(4x8)

32 1 2 3 4 5
 4 5 6 7 8 9
 8 9 10 11 12 13
 12 13 14 15 16 17
 16 17 18 19 20 21
 20 21 22 23 24 25
 24 25 26 27 28 29
 28 29 30 31 32 1

1 0 0 0 0 0
 0 0 0 0 0 0
 0 0 0 0 0 0
 0 0 0 1 1 0
 1 0 0 0 0 0
 0 0 0 0 1 0
 1 0 0 0 0 0
 0 0 0 0 1 0

→ 48 bits
 (6x8)

Step-9 Input ⊕ key.

1 0 0 0 0 0 0 0	⊕	0 0 0 0 1 1	1 0 0 0 1 1
0 0 0 0 0 0 0 0		0 0 0 0 1 1	0 0 0 0 1 1
0 0 0 0 0 0 0 0		1 0 0 1 1 0	1 0 0 1 1 0
0 0 0 1 1 0		0 0 1 1 0 1	0 0 1 0 1 1
1 0 0 0 0 0		1 0 0 0 1 1	0 0 0 0 1 1
0 0 0 0 1 0		1 0 0 0 0 1	1 0 0 0 1 1
1 0 0 0 0 0		0 0 1 0 0 1	1 0 1 0 0 1
0 0 0 0 1 0		1 1 1 1 0 0	1 1 1 1 1 0

└───┘

S-BOX

(15)

S-Box for Block-2

① 100011

R = 11 2 3 output = 12
C = 0001 2 1 binary = 1100

② 000011

; ③ 100110

R = 01 2 1 0 = B ; R = 10 2 2 0 = 09
C = 0001 2 1 B = 1101 ; C = 0011 2 3 B = 1001

④ 001011

R = 01 2 1 0 = 15
C = 0101 2 5 B = 1111

⑤ 000011

R = 01 2 1 0 = 11
C = 01 2 1 B = 1011

⑥ 100011

R = 11 2 3 0 = 03
C = 0001 2 1 B = 0011

⑦ 101001

R = 3 2 11 0 = 01
C = 4 2 0100 B = 0001

⑧ 111110

R = 10 2 2 0 = 08
C = 1111 2 15 B = 1000

⑨

Step-11 Permutation

4	1	1	0	0	16	07	20	21	29	12	28	17
8	1	1	0	1	01	15	23	26	05	18	31	10
12	1	0	0	1	02	08	24	14	32	27	03	09
16	1	1	1	1	19	13	30	06	22	11	04	25
20	1	0	1	1								
24	0	0	1	1								
28	0	0	0	1								
32	1	0	0	0								

Straight Permutation Box

S-Box
Values

Output

```

1 0 1 0 1 1 1 1
1 1 1 0 1 0 0 0
1 1 1 1 0 0 0 0
1 1 0 1 0 0 0 1

```

Step 12 XOR with first half

10101111		000000001
11101000	(+)	000000000
11110000		000000000
11010001		000000011

Output

```

10101110
11101000
11110000
11010010

```

Final answer after Round 1 of Second Block.

	ASCII	Symbol
000000001	01	SOH
000000000	00	NUL
000000000	00	NUL
000000011	03	ETX
10101110	174	@
11101000	232	è
11110000	240	ð
11010010	210	ð

Final answer (Cipher) Block 1 + Block 2

NULpO<<D"êSOHNULNULETX@èðð

Cipher of PT

Cipher of Padkey

Q.2

RSA

20

$$M = 3$$

$$P = 19, \quad q = 13$$

Two Prime numbers

$$n = P \times q = 19 \times 13 = 247$$

$$\phi(n) = (P-1)(q-1) = (18)(12) = 216$$

For e

Condition for calculating 'e'

$$(1) \quad 1 < e < \phi(n)$$

$$(2) \quad \gcd(e, \phi(n)) = 1$$

e should be that value whose greatest common factor of $\phi(n) = 1$

They must be COPRIME

$$e < 10, \quad \phi(n) = 216$$

$$\gcd(1, 216) = 1$$

$$\gcd(2, 216) = 2$$

$$\gcd(3, 216) = 3$$

$$\gcd(4, 216) = 4$$

$$\gcd(5, 216) = 1$$

$$\gcd(6, 216) = 6$$

$$\gcd(7, 216) = 1$$

$$\gcd(8, 216) = 8$$

$$\gcd(9, 216) = 9$$

We have now 2 valid options for e whose $\gcd$ with $\phi(n)$ produce/is 1. But we need to select that value of e by which we can obtain value of d to successfully decrypt the cipher.

For d

$$d \cdot e \equiv 1 \pmod{\phi(n)}$$

$e = 5$

or

$$e \cdot d \pmod{\phi(n)} = 1$$

By which we get,

$$d \equiv e^{-1} \pmod{\phi(n)}$$

method 1

$$d = \frac{(\phi(n) \times i) + 1}{e}$$

method 2

Method 1

$$d = 5^{-1} \pmod{216}$$

$\therefore$ adding 1 to $\phi(n)$

$$1 + 216 = 433 = 649 = 865$$

$$\begin{array}{cc} & \swarrow \searrow \\ 5 & 173 \end{array}$$

$$\text{So, } 5 \times 173 = 865$$

which gives 173 as modulo inverse of 5

$$\boxed{d = 173}$$

Method 2

$$d = \frac{(\phi(n) \times i) + 1}{e} = \frac{(216 \times 1) + 1}{5}$$

$$d = \frac{217}{5} = 43, \text{ we need modulo, so subtract it from } \phi(n)$$

$$= 216 - 43 = \boxed{173}$$

Both methods can be used

For d

$$e = 7$$

$$d \cdot e \equiv 1 \pmod{\phi(n)}$$

$$e \cdot d \pmod{\phi(n)} = 1 \Rightarrow d \equiv e^{-1} \pmod{\phi(n)}$$

$$d \equiv 7^{-1} \pmod{216}$$

$$1 = 217$$

$$\begin{array}{c} 1 \\ \swarrow \searrow \\ 7 \quad 31 \end{array}$$

So

7 x 31 give 217, so

31 is the modular inverse of 7

$$\boxed{d = 31}$$

By formula

$$d \equiv (e^{-1} \pmod{\phi(n)}) + 1$$

$\therefore 1 = \text{initial value}$

$$d \equiv \frac{(216 \times 1) + 1}{7}$$

$$\boxed{d = 31}$$

For encryption

$$e = 7$$

$$C \equiv M^e \pmod{n}$$

$$C \equiv (3)^7 \pmod{247}$$

$$C \equiv 243 \pmod{247}$$

$$\boxed{C = 243}$$

$$e = 7$$

$$C \equiv M^e \pmod{247}$$

$$C \equiv 3^7 \pmod{247}$$

$$C \equiv 2187 \pmod{247}$$

$$\boxed{C = 211}$$